

SEA++: Multi-Graph-Based Higher-Order Sensor Alignment for Multivariate Time-Series Unsupervised Domain Adaptation

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Abstract—Unsupervised Domain Adaptation (UDA) methods have been successful in reducing label dependency by minimizing the domain discrepancy between labeled source domains and unlabeled target domains. However, these methods face challenges when dealing with Multivariate Time-Series (MTS) data. MTS data typically originates from multiple sensors, each with its unique distribution. This property poses difficulties in adapting existing UDA techniques, which mainly focus on aligning global features while overlooking the distribution discrepancies at the sensor level, thus limiting their effectiveness for MTS data. To address this issue, a practical domain adaptation scenario is formulated as Multivariate Time-Series Unsupervised Domain Adaptation (MTS-UDA). In this paper, we propose Sensor Alignment (SEA) for MTS-UDA, aiming to address domain discrepancy at both local and global sensor levels. At the local sensor level, we design endo-feature alignment, which aligns sensor features and their correlations across domains. To reduce domain discrepancy at the global sensor level, we design exo-feature alignment that enforces restrictions on global sensor features. We further extend SEA to SEA++ by enhancing the

endo-feature alignment. Particularly, we incorporate multi-graph-based higher-order alignment for both sensor features and their correlations. Extensive empirical results have demonstrated the state-of-the-art performance of our SEA and SEA++ on six public MTS datasets for MTS-UDA.

Index Terms—Multivariate time-series data, unsupervised domain adaptation, graph neural network.

I. INTRODUCTION

TIME-SERIES (TS) data has gained significant attention in various fields due to their wide range of applications. To address TS-related problems, Deep Learning (DL) methods have played a pivotal role in capturing latent dependencies within TS data [1], [2], [3], [4], [5], [6], [7]. However, the effectiveness of DL models heavily relies on the availability of large-scale labeled data, which is often a challenge in real-world applications due to the high cost associated with data labeling.

To reduce the cost of labeling, Unsupervised Domain Adaptation (UDA) methods have been proposed to transfer the knowledge from a labeled source domain to an unlabeled target domain, enabling effective learning when target domain labels are unavailable [8]. Due to domain shifts, existing UDA methods focused on reducing discrepancy across domains by learning domain-invariant features [9], [10], [11], [12], [13], [14]. These methods have achieved decent performance, showing the effectiveness of UDA in reducing label dependency. Subsequently, researchers applied UDA to TS data, aiming to reduce domain discrepancy by learning domain-invariant temporal features extracted by Recurrent Neural Network (RNN) [15], Long Short-Term Memory (LSTM) [16], and Convolutional Neural Network (CNN) [17].

While substantial progress has been made, existing UDA methods may be inapplicable for scenarios where multiple sensors are deployed simultaneously. For instance, in Remaining Useful Life (RUL) prediction, various types of sensors measure different physical parameters, while in Human Activity Recognition (HAR), sensors positioned at various locations on the human body capture activity data. We formulate these scenarios as Multivariate Time-Series Unsupervised Domain Adaptation (MTS-UDA), a challenging yet practical problem. Applying current TS UDA methods directly to MTS-UDA poses two key limitations. First, MTS data inherently exhibits a multi-source

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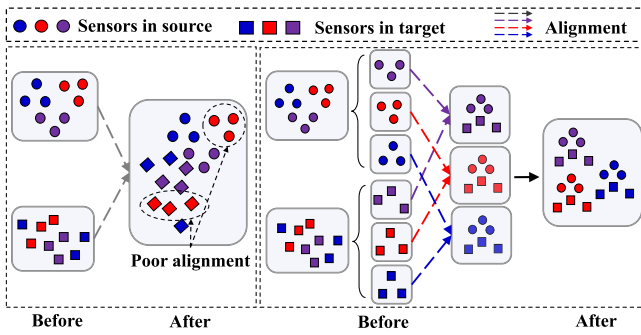


Fig. 1. Comparisons between existing UDA methods and ours, where before and after represent before and after alignment respectively. Left: Only global sensor features are aligned, resulting in the poor alignment of the red sensor. Right: SEA aligns the sensor information between domains, so the feature discrepancy of each sensor can be reduced.

nature, with each data typically originating from multiple sensors. Each sensor captures data with its unique distribution due to differences in sensor types or deployment locations [18]. When adapting existing TS UDA methods to MTS-UDA, they primarily focus on aligning global features across domains by treating signals from various sensors as a whole [19]. However, these approaches ignore the diverse distributions across sensors, leading to misalignment at the sensor level and consequently suboptimal solutions for MTS-UDA. As presented in Fig. 1 (Left), existing UDA methods focus on aligning global features between domains, leading to inadequate alignment of features from the red sensor and thereby limiting the model's transferability. Second, MTS data contains essential spatial-temporal dependency information due to its multi-source nature. Specifically, the spatial dependency refers to the correlations between sensors, capturing the critical interactive sensor relationships. For instance, the spatial dependency in machine RUL prediction corresponds to the correlation between a temperature sensor and a fan speed sensor, as they exhibit strong correlations. Besides, the temporal dependency refers to the temporal relationships between consecutive timestamps. Existing UDA methods are limited in effectively modelling and transferring both dependencies across domains, thus leading to suboptimal solutions for MTS-UDA. These properties of MTS-UDA render it a more complex and challenging task than standard TS UDA.

To address the above challenges, we propose Sensor Alignment (SEA) for MTS-UDA. First, we aim to reduce domain discrepancy at both the local and global sensor levels. At the local sensor level, we propose endo-feature alignment to align sensor-level information between domains. This alignment process includes aligning both sensor features and sensor correlations, which capture the information of individual sensors and their interactions, respectively. To achieve this alignment, we introduce contrastive sensor alignment and sensor correlation alignment. Further, at the global sensor level, we design exo-feature alignment to reduce global feature discrepancy. This is achieved by enforcing restrictions on the global features derived from stacked sensor features. Moreover, to capture the

spatial-temporal dependencies inherent in MTS data and enable their effective transfer across domains, we introduce a graph-based framework that can capture the dependency information. Given that MTS data may not come with pre-defined graphs representing the relationships between sensors, we propose a multi-branch self-attention mechanism to model these dependencies.

Although SEA presents a feasible solution for sensor alignment in MTS-UDA, it can be further improved from two perspectives. First, SEA employs a lower-order statistics-based scheme for endo-feature alignment, which may be insufficient to achieve comprehensive alignment in scenarios with complex data distributions. Second, SEA can benefit from leveraging the dynamically evolving distributions within MTS data to enhance its performance. For instance, as a machine's health deteriorates, the distribution of a fan speed sensor undergoes dynamic changes due to increased friction. Taking these issues into consideration, we develop SEA++ as an extension of SEA by introducing multi-graph-based higher-order alignment to enhance our endo-feature alignment, incorporating improved sensor correlation and contrastive sensor alignment. To achieve comprehensive sensor alignment, we incorporate higher-order statistics by capturing complex data distributions across domains. Additionally, SEA++ introduces a multi-graph alignment technique tailored for aligning the evolving distributions within MTS data. This involves constructing sequential graphs that represent the distributions within local temporal intervals, followed by the alignment of corresponding sequential graphs between domains. To optimize the alignment process, weights for balancing sequential graphs are learned based on distribution discrepancies, enabling prioritized alignment for challenging graphs. This module effectively aligns the evolving distributions, enabling adaptation to the evolving nature of MTS data. With the two types of enhancements in the improved version, we represent the name with two plus signs as SEA++.

Our contributions can be summarized as following.

- We analyze the challenging Multivariate Time-Series Unsupervised Domain Adaptation (MTS-UDA) in accordance with the inherent nature of MTS data and propose Sensor Alignment (SEA) that reduces domain discrepancy at both the local and global sensor levels. Meanwhile, SEA captures the spatial-temporal dependencies within MTS data, enabling effective transfer of the dependency information across domains.
- To cope with scenarios with complex data distributions, we introduce SEA++ by enhancing our endo-feature alignment. Higher-order statistics alignment methods are employed to achieve comprehensive sensor correlation and sensor feature alignment.
- Considering the dynamically evolving distributions within MTS data, SEA++ further incorporates a multi-graph alignment technique. It aligns sequential graphs representing local distributions, effectively aligning evolving distributions between domains.
- We evaluate the effectiveness of SEA and SEA++ with six various real-world MTS datasets through extensive

experiments, demonstrating that both methods achieve state-of-the-art performance for MTS-UDA.

This journal paper presents an extended version of our previous work [20] by introducing multi-graph-based higher-order alignment techniques. First, we introduce higher-order statistic alignment to improve endo-feature alignment, aiming to achieve more comprehensive sensor-level alignment. Second, we analyze the properties of MTS data and emphasize the significance of dynamically evolving distributions inherent within MTS data. Thus, we introduce a multi-graph alignment technique to align the evolving distributions across domains. Third, we have conducted additional experiments, including more datasets, more compared methods, more comprehensive ablation studies, and extensive sensitivity analysis, to fully assess SEA and SEA++.

II. RELATED WORK

A. Unsupervised Domain Adaptation

UDA aims to mitigate the need for labeled data by transferring knowledge from a labeled source domain to an unlabeled target domain. To achieve good performance on the target domain, existing UDA methods mainly focus on minimizing domain discrepancies by utilizing two categories of approaches. The first category comprises adversarial-based methods, which utilize domain discriminator networks to encourage a feature extractor to learn domain-invariant representations. Prominent models include Domain Adversarial Neural Network (DANN) [10] and Adversarial Discriminative Domain Adaptation (ADDA) [11]. These models have served as foundations for various improved variants to address specific challenges [12], [21], [22], [23], [24], [25]. For instance, CDAN [12] and MADA [21] were both extensions of DANN to incorporate multi-mode structure information. DAAN [22] introduced class-aware adversarial loss to align the conditional distribution using DANN. DIRT-T [26] improved upon existing adversarial-based methods by integrating adversarial training with a penalty term to punish violations of the cluster assumption.

The second category encompasses metric-based methods which enforce metric restrictions to learn invariant features [9], [13], [27], [28]. One pioneering method is Maximum Mean Discrepancy (MMD) [9]. JAN [14] was developed based on MMD to align the joint distributions of feature and label distributions. MK-MMD [29] employed MMD to align features in the final layers, enhancing their transferability. To achieve comprehensive alignment across domains, higher-order statistical alignment methods have been proposed. Deep CORrelation ALignment (CORAL) [13] emerged as a standard approach introducing second-order alignment. Its effectiveness has inspired subsequent works aiming to align higher-order statistics across domains [30], [31], [32], [33]. Specifically, MMCD [31], [32] combined MMD with a maximum covariance discrepancy to explore second-order statistics in reproducing kernel Hilbert space. Rahman et al. [30] simultaneously aligned the first and second-order statistics by utilizing MMD and Deep CORAL. Furthermore, HoMM [34] was developed as a third-order metric technique to align feature distributions in hidden layers. CMD [33] was proposed to match the higher-order central

moments of probability distributions by means of order-wise moment differences. Overall, these methods have contributed to the advancement of UDA by effectively reducing domain discrepancy and enabling knowledge transfer across domains.

B. Unsupervised Domain Adaptation for Time-Series Data

In recent years, there has been a growing trend of exploring UDA methods in various domains, showing their effectiveness in different applications [35], [36]. In TS data, researchers primarily focused on reducing domain discrepancy by aligning temporal features across domains [15], [16], [17], [19], [37]. Several notable works have made significant contributions to this area. For instance, VRADA [15] leveraged variational RNN to learn temporal features and employed adversarial-based methods to reduce domain discrepancy. Zhao et al. [38] combined DANN and CNN for TS classification, incorporating a center loss to minimize intra-class variation and maximize inter-class distance. CoDATS [39] and CLADA [40], as multi-source TS UDA methods, applied DANN and 1D-CNN to handle the scenarios with multiple source domains. Moreover, researchers have designed UDA methods specifically tailored to TS data. ADATIME [19] considered TS temporal properties and evaluated various temporal encoders for TS UDA. AdvSKM [41] mapped temporal features into a high-dimensional space with adversarial Spectral Kernels and then aligned them using MMD. SASA [16] aligned the associative structure between variables to achieve invariant structural information for TS UDA. CoT-Mix [42] adopted a contrastive loss to reduce domain distribution shifts without relying on adversarial or metric-based methods. Felix et al. [43] searched for an optimal transformation to convert input data from a source domain to a target domain for domain-invariant representations.

These works are great pioneers for TS UDA, but they may not perform well for the scenarios requiring multiple sensors, i.e., MTS data. In MTS data, signals from different sensors often follow various distributions due to their varied physical locations and measurements. Existing TS UDA methods can be adapted to MTS data by treating all sensors as a whole [15], [19], [38], [44], facilitate the learning of global features for alignment, but they are limited in accounting for the distinct distributions of sensors within MTS data. Meanwhile, they ignore the multi-source nature of MTS data, restricting their ability to model and transfer spatial-temporal dependency information across domains. To address these limitations, we propose our SEA and SEA++ that accounts for aligning sensor-level information and modelling dependency information within MTS data to achieve better MTS-UDA.

III. SEA MODEL

A. Problem Definition

For MTS-UDA, we are given a source domain with N_s labeled samples $\mathcal{D}_S = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and a target domain with N_t unlabeled samples $\mathcal{D}_T = \{x_i^t\}_{i=1}^{N_t}$. Each MTS sample x_i (either x_i^s or x_i^t) originates from N sensors with different distributions, i.e., $x_i = \{x_{im}\}_{m=1}^N \in \mathbb{R}^{N \times L}$, where L represents the time

length. The goal of SEA is to train an encoder by transferring the knowledge from the source domain to the target domain, enabling effective feature extractions $h_i \in \mathbb{R}^F$ from MTS data x_i^t in the target domain. The features can then be used for downstream tasks. Notably, the SEA framework consists of two stages. In the first stage, we focus on feature extraction from MTS data. To simplify the notation and enhance clarity, we omit the index i in this stage. Thus, each sample is represented as x^s and x^t , and the data from the m th sensor is denoted as x_m^s and x_m^t . While in the second stage that involves domain alignment, we retain the use of the index to facilitate a clear description of the alignment process.

Furthermore, to model the evolving distributions within MTS data, we construct sequential graphs $\{\mathcal{G}_T\}_{T=1}^{\hat{L}}, \mathcal{G}_T = (Z_T, E_T)$ from the sample x , where $Z_T = \{z_{m,T}\}_{m=1}^N$ and $E_T = \{e_{mn,T}\}_{m,n=1}^N$ represent sensor features and correlations, respectively, in the T th graph. $\{\mathcal{G}_T^s\}_{T=1}^{\hat{L}}$ and $\{\mathcal{G}_T^t\}_{T=1}^{\hat{L}}$ represent the sequential graphs in the source and target domains respectively.

B. Preliminary

1) *Graph Neural Network*: Several previous studies have demonstrated the effectiveness of Graph Neural Network (GNN) in capturing spatial dependencies [45], [46], [47]. Motivated by this capability, we employ GNN to capture the spatial dependencies, i.e., sensor correlations, within our constructed sequential graphs. Specifically, we utilize a variant of GNN known as Message Passing Neural Network (MPNN) [48] to process each graph. The MPNN framework consists of two stages: propagation and updating. During the propagation stage, the features of neighboring nodes are propagated to the central node. Subsequently, the aggregated features are updated using a non-linear function, such as a Multi-Layer Perceptron (MLP) network. With MPNN, we can capture the spatial dependencies between sensors, enabling the effective transfer of the dependency information across domains in the subsequent alignment processes. The details of MPNN are shown in (1), where sensor m is the central node, and $\mathcal{N}(m)$ is the set of its neighboring nodes.

$$\begin{aligned} h_{m,T} &= \sum_{j \in \mathcal{N}(m)} e_{mj,T} z_{j,T}, \\ z_{m,T} &= \text{ReLU}(h_{m,T} W_G). \end{aligned} \quad (1)$$

2) *Deep CORAL*: Deep CORAL [13] is a typical metric-based UDA method. Different from $\mathcal{L}_2$ or cosine distances mainly employing first-order statistics, Deep CORAL focuses on minimizing the discrepancy in second-order statistics between domains [49], making it possible to capture complex data distributions for alignment. Given training batches with n^s and n^t samples in source and target domains respectively, we denote the extracted features from these domains as $\mathcal{H}^s \in \mathbb{R}^{n^s \times f}$ and $\mathcal{H}^t \in \mathbb{R}^{n^t \times f}$, where f represents the feature dimension. With the features, we can compute the discrepancy of second-order statistics with Deep CORAL $M_c(\mathcal{H}^s, \mathcal{H}^t)$ as (2), where $\|\cdot\|_F^2$ is the squared matrix Frobenius norm. By minimizing the Deep

CORAL discrepancy $\mathcal{L}_{\text{CORAL}}$, the second-order statistics of the features are aligned between source and target domains, leading to comprehensive alignment across domains.

$$\begin{aligned} C^s &= \frac{(\mathcal{H}^s)^T \mathcal{H}^s - \frac{1}{n^s} (\mathbf{1}^T \mathcal{H}^s) (\mathbf{1}^T \mathcal{H}^s)}{n^s - 1}, \\ C^t &= \frac{(\mathcal{H}^t)^T \mathcal{H}^t - \frac{1}{n^t} (\mathbf{1}^T \mathcal{H}^t) (\mathbf{1}^T \mathcal{H}^t)}{n^t - 1}, \\ M_c(\mathcal{H}^s, \mathcal{H}^t) &= \frac{1}{4f^2} \|C^s - C^t\|_F^2. \end{aligned} \quad (2)$$

C. Overall Structure

The overall structure of SEA is shown in Fig. 2. The superscripts s and t are omitted in Section III-C and III-D, as the forward processes in source and target domains share the same graph-based encoder, including Multi-branch Self-attention based Graph Construction (MSGC), GNN, and LSTM. First, we model the spatial-temporal dependencies within MTS data for effective transfer across domains. To capture the evolving distributions within MTS data, we segment each sample into multiple patches, constructing sequential graphs by MSGC. Then, leveraging these sequential graphs, we utilize GNN and LSTM to capture the spatial-temporal information for learning sensor-level information, i.e., sensor features and correlations. Second, in addition to supervised learning on the source domain, we incorporate endo-feature alignment and exo-feature alignment techniques to reduce domain discrepancy at both local and global sensor levels. At the local sensor level, we design endo-feature alignment, aligning sensor features and sensor correlations by proposing Sensor Feature Alignment (SFA) and Sensor Correlation Alignment (SCA) respectively. The two components are further improved by both higher-order statistic alignment and multi-graph alignment to form iSFA and iSCA. The higher-order statistic alignment aims to achieve comprehensive alignment by capturing complex data distributions, while the multi-graph alignment aims to align the evolving distributions across domains at the sensor level. We introduce the details of each module in subsequent sections.

D. Graph-Based Encoder

1) *Segmentation*: To capture the evolving distributions within MTS data, we propose segmenting each sample as multiple patches with a predefined patch size, where each patch represents the distribution in local temporal intervals. As shown in Fig. 3, a sample $x \in \mathbb{R}^{N \times L}$ is segmented into multiple patches with a predefined size d , yielding sequential features $\{Z_T\}_{T=1}^{\hat{L}} \in \mathbb{R}^{N \times \hat{L} \times d}$. Here, $\hat{L} = \lfloor \frac{L}{d} \rfloor$ represents the number of patches, where $\lfloor \cdot \rfloor$ denotes the truncation operation. With each patch, the features $Z_T \in \mathbb{R}^{N \times d}$ capture the evolving distributions in local temporal intervals. Specifically, the features of the m th sensor are denoted as $z_{m,T} \in \mathbb{R}^d$. Furthermore, we incorporate a nonlinear function $f_L(\cdot)$ to enhance the nonlinear expressiveness of the model.

2) *MSGC*: We construct sequential graphs with the sequential features. Traditionally, various methods were employed to

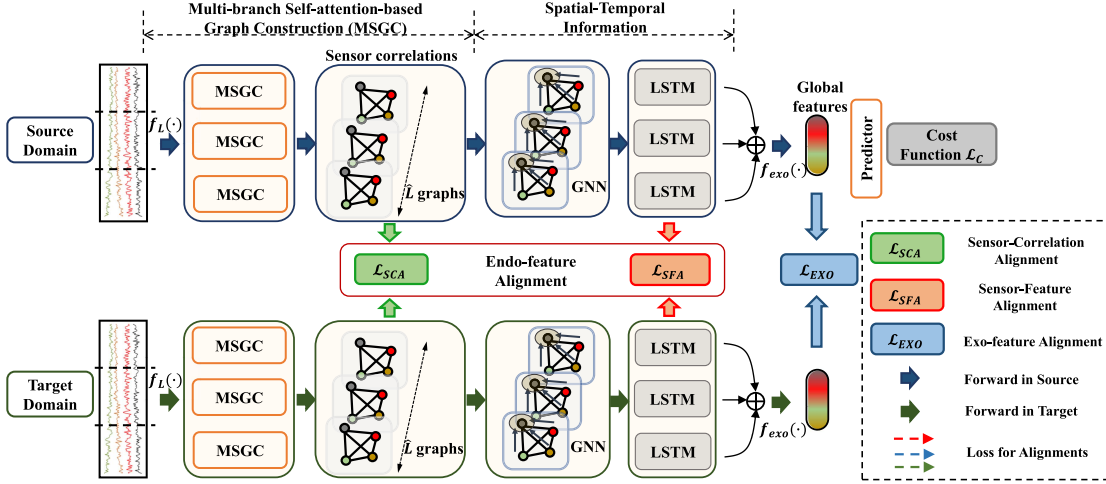


Fig. 2. The overall structure. (1) To transfer spatial-temporal dependencies across domains, source and target domains share the same graph-based encoder, including MSGC, GNN, and LSTM. Each sample is segmented as multiple patches, which are constructed as sequential graphs. With the combination of GNN and LSTM, decent sensor information is learned, including sensor features and correlations. (2) In addition to supervised learning on the source domain, endo-feature alignment and exo-feature alignment are designed to reduce domain discrepancy at the local and global sensor levels. The endo-feature alignment includes aligning sensor features and their correlations between domains by SCA and SFA, both of which are enhanced by incorporating multi-graph alignment and higher-order statistic alignment. Exo-feature alignment aligns global features mapped by sensor features.

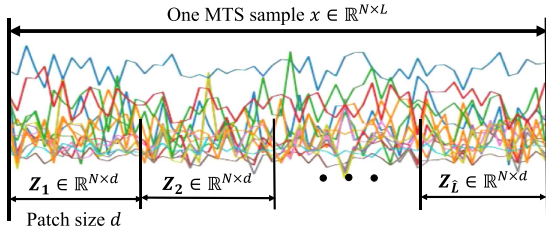


Fig. 3. Diagram of segmenting each MTS sample into patches.

construct graphs for MTS data, such as dot-product [50], Euclidean distance [46], and concatenation [51]. However, the traditional methods struggle to model comprehensive relationships among sensors due to the complex topological structures presented within MTS data. To address the limitation, we propose MSGC, which constructs a graph by combining multiple graphs generated from various weight initializations across multiple branches. Each branch employs differently weighted matrices to model sensor correlations from distinct perspectives, enabling us to capture comprehensive sensor correlations. Specifically, we adopt query and key to learn the edge weights between sensors. Equation (3) describes the edge between sensors m and n in the i th branch.

$$\begin{aligned} q_{m,T}^i &= z_{m,T} W_Q^i, k_{n,T}^i = z_{n,T} W_K^i, \\ e_{mn,T}^i &= \frac{\exp(\frac{(q_{m,T}^i)^T k_{n,T}^i}{\sqrt{d}})}{\sum_{j=1}^N \exp(\frac{(q_{m,T}^i)^T k_{j,T}^i}{\sqrt{d}})}, \end{aligned} \quad (3)$$

where $z_{m,T}$ and $z_{n,T}$ represent the query and key sensors, respectively, to learn the edge weight between them for the i th branch, i.e., $e_{mn,T}^i$. The edge weights are then normalized using the softmax function to ensure the weight within $[0, 1]$.

Then, we combine the graphs from multiple branches to capture comprehensive sensor correlations. Specifically, we calculate the average of corresponding edges: $e_{mn,T} = \sum_{i=1}^{n_b} \frac{e_{mn,T}^i}{n_b}$, where n_b represents the number of branches, resulting in $E_T = \{e_{mn,T}\}_{m,n=1}^N \in \mathbb{R}^{N \times N}$ for the T th graph to represent sensor correlations. By conducting similar operations in all patches, we can obtain sequential correlations $\{E_T\}_{T=1}^{\tilde{L}} \in \mathbb{R}^{\tilde{L} \times N \times N}$. The difference between MSGC with graph attention networks can be found in our appendix, available online.

3) *Capturing Spatial-Temporal Information*: With the sequential graphs, we then capture the spatial-temporal dependencies within MTS data. First, we utilize MPNN in (1) to capture spatial dependencies within each sequential graph. In this stage, each graph is processed with shared weights to reduce trainable parameters. After MPNN updates sensor features in sequential graphs, we then capture temporal dependencies. Notably, features from corresponding sensors across sequential graphs exhibit temporal dependencies. Leveraging its robust capability to capture such dependencies, we employ LSTM to capture the dependency information for each sensor. Specifically, the features $z_m \in \mathbb{R}^{\tilde{L} \times d}$ from each sensor across sequential graphs are feeded into LSTM. After GNN and LSTM, we finally obtain updated sequential features $\{Z_T\}_{T=1}^{\tilde{L}} \in \mathbb{R}^{\tilde{L} \times N \times d}$. By combining the capabilities of the two networks, we can effectively capture the spatial-temporal dependencies within MTS data, allowing to transfer the dependency information across domains through the alignment at the local and global sensor levels in the subsequent stages.

E. Sensor Alignment

To reduce the discrepancy between domains for MTS-UDA, we design endo-feature alignment and exo-feature alignment at the local and global sensor levels respectively.

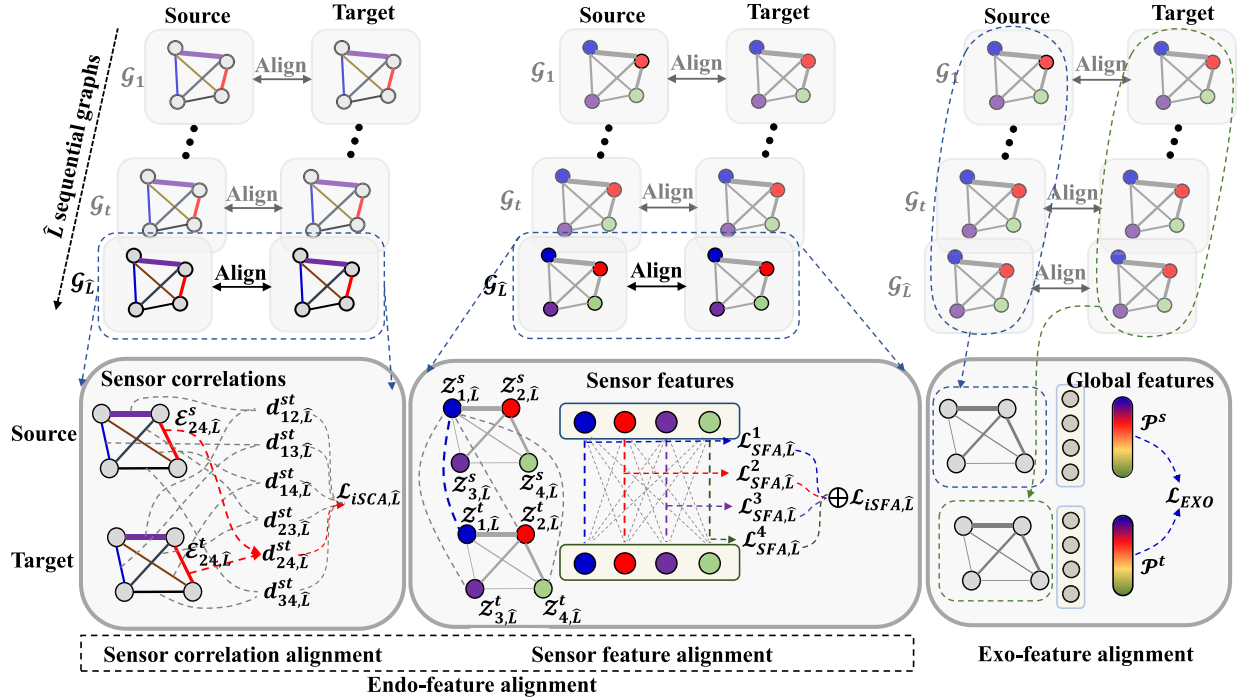


Fig. 4. Left: Sensor correlation alignment is to make the correlations between domains identical. Middle: Sensor feature alignment is to make the features from the corresponding sensors in two domains similar and the data from the different sensors in two domains different. Right: Exo-feature alignment reduces the discrepancy between the global features in different domains.

1) *Endo-Feature Alignment*: Existing UDA methods reduce domain discrepancy by aligning global features across domains, which is applicable for the data coming from a single source. However, MTS data originate from multiple sensors, with each sensor following various distributions. When adapting existing UDA methods to MTS-UDA, they can consider the global distribution by treating all sensors as a whole yet ignoring the distinct distributions of each sensor. This potentially results in the misalignment at the sensor level, restricting the performance of transferring sensor-level information from the source domain to the target domain. To address this limitation, we propose an endo-feature alignment method designed to align sensor-level information and avoid misalignment for each sensor when transferring knowledge across domains.

To make better endo-feature alignment, it's crucial to note that sensor-level information comprises both sensor features and their correlations. Sensor features describe each sensor's information, e.g., its unique distribution, while sensor correlations capture interactive information among sensors. For instance, we present a scenario with temperature and fan speed sensors deployed together to monitor a machine's status. It's desirable for temperature sensor data in two domains to exhibit similar distributions. Additionally, the correlation between the sensors should consistently reflect patterns across domains, such as the fan speed consistently increasing with temperature rises. Based on these considerations, aligning sensor features and sensor correlations across domains simultaneously is important to ensure comprehensive sensor-level alignment.

a) *Sensor Correlation Alignment*: To explore consistent patterns across domains for sensor interactions, we propose to

align the correlations among sensors. Specifically, we align the distributions of correlations between any two sensors across domains. To achieve this, we propose sensor correlation alignment (SCA), which involves individually aligning each edge of the graphs as shown in Fig. 4 Left.

Given training batches with n^s and n^t samples in source and target domains, we represent sensor correlations across sequential graphs as $\mathcal{E}^s \in \mathbb{R}^{n^s \times \hat{L} \times N \times N}$ and $\mathcal{E}^t \in \mathbb{R}^{n^t \times \hat{L} \times N \times N}$ respectively, with $\mathcal{E}_T^s \in \mathbb{R}^{n^s \times N \times N}$ and $\mathcal{E}_T^t \in \mathbb{R}^{n^t \times N \times N}$ signifying sensor correlations for the T th graph. Here, $\mathcal{E}_T^s = \{\mathcal{E}_{mn,T}^s\}_{m,n=1}^N$, wherein $\mathcal{E}_{mn,T}^s = \{e_{i,mn,T}^s\}_i^{n^s}$. For SCA, the alignment across domains is achieved based on averaged sensor correlations. Specifically, for the edge connecting sensors m and n , its mean value over a training batch is $e_{mn,T}^s = \sum_i^{n^s} e_{i,mn,T}^s / n^s$. Subsequently, the averaged sensor correlations over sequential graphs are calculated as $e_{mn}^s = \sum_T^{\hat{L}} e_{mn,T}^s / \hat{L}$. Next, we minimize the difference across domains for averaged sensor correlations, as shown in (4), where $|\cdot|$ is the absolute value. This involves minimizing the expected discrepancy of sensor correlations across domains. By doing so, we can reduce distribution discrepancies for sensor correlations in source and target domains.

$$e_{mn}^{st} = e_{mn}^s - e_{mn}^t, \\ \mathcal{L}_{SCA} = \mathbb{E}(|e_{mn}^{st}|). \quad (4)$$

b) *Sensor Feature Alignment*: To align sensor features between domains, we assume that features originating from corresponding sensors in two domains should exhibit similar

distributions. Meanwhile, compared with features from corresponding sensors, features from different sensors in the two domains should have distinct distributions. For example, as shown in Fig. 4 **Middle**, the distribution of sensor 1 in the source domain should be more similar to the distribution of sensor 1 in the target domain than other sensors, e.g., 2, 3, and 4. Thus, we propose a sensor contrastive mechanism to facilitate the alignment of sensor features across domains.

Given training batches with n^s and n^t samples, we define sensor features as $\mathcal{Z}^s \in \mathbb{R}^{n^s \times \hat{L} \times N \times d}$ and $\mathcal{Z}^t \in \mathbb{R}^{n^t \times \hat{L} \times N \times d}$, with $\mathcal{Z}_T^s \in \mathbb{R}^{n^s \times N \times d}$ and $\mathcal{Z}_T^t \in \mathbb{R}^{n^t \times N \times d}$ denoting sensor features for the T th graph. Here, $\mathcal{Z}_T^s = \{\mathcal{Z}_{n,T}^s\}_{n=1}^N$, wherein $\mathcal{Z}_{n,T}^s = \{z_{i,n,T}^s\}_i^{n^s}$. In SFA, domain contrasting is achieved based on averaged sensor features. In (5), p_m^s represents the averaged sensor features over multiple graphs within a training batch, calculated as $p_m^s = \sum_T \sum_i z_{i,n,T}^s / n^s \hat{L}$.

$$\mathcal{L}_{SFA} = -\frac{1}{N} \sum_n \log \frac{e^{\varphi(p_m^s, p_m^t)}}{\sum_{p_j^t \in P^t} e^{\varphi(p_m^s, p_j^t)}},$$

$$\varphi(p_m^s, p_m^t) = p_m^s (p_m^t)^T. \quad (5)$$

c) *Exo-Feature Alignment*: To reduce domain discrepancy at the global level, we enforce restrictions on the global features mapped by sensor features. We first learn global features for each sample by stacking all sensor features across sequential graphs and employing an MLP network to map the stacked features as $\mathcal{P}^s$ and $\mathcal{P}^t$ for two domains, as shown in Fig. 4 **Right**. With the global features, we then align with Deep CORAL, expressed as $\mathcal{L} = \mathcal{M}_c(\mathcal{P}^s, \mathcal{P}^t)$. Finally, the overall loss function (6) is minimized:

$$\min \mathcal{L} = \mathcal{L}_C + \mathcal{L}_{EXO} + \mathcal{L}_{Endo},$$

$$\mathcal{L}_{Endo} = \lambda_{SCA} \mathcal{L}_{SCA} + \lambda_{SFA} \mathcal{L}_{SFA}. \quad (6)$$

where $\mathcal{L}_C$ is the cost function (such as mean square error loss or cross-entropy loss, depending on specific tasks) computed by the source data and labels, and λ_{SCA} and λ_{SFA} are hyperparameters to balance the effect of SCA and SFA respectively.

IV. SEA++ MODEL

While the method outlined above provides a feasible strategy for sensor alignment, there are still potential improvements to enhance endo-feature alignment. Two key directions warrant consideration. First, SEA employs first-order statistics for alignment, such as averaged sensor features over a training batch used in sensor contrastive mechanism. However, the first-order alignment is limited in dealing with complex data distributions, restricting its ability to achieve comprehensive alignment across domains and yielding suboptimal performance. To address that, we propose introducing second-order statistical metrics for alignment to achieve more comprehensive endo-feature alignment.

Second, the constructed sequential graphs, comprising sequential features and correlations, capture evolving distributions within local temporal intervals and should be aligned across domains. We consider the example of monitoring machine health

status for illustration. As the machine's health deteriorates, the distribution of a fan speed sensor may change due to factors like increased frictions. Similarly, the interaction between the fan speed sensor and temperature sensors will also be affected and undergo changes. Specifically, the same fan speeds might lead to higher temperature increases due to a machine in poor health, such as increased friction. Thus, we emphasize the importance of aligning the evolving distributions between domains at the sensor level. However, SEA employs averaged sensor features and correlations over sequential graphs, limiting its ability to align the evolving distributions within MTS data. To improve this, we introduce a multi-graph alignment technique, aiming to effectively align evolving distributions between domains.

Next, we begin by discussing SCA and SFA enhanced by higher-order statistical alignment within each sequential graph. Subsequently, we focus on combining the alignment across all sequential graphs by employing our multi-graph alignment.

A. Sensor Correlation Alignment

Given training batches with n^s and n^t samples in source and target domains, we have sensor correlations for the T th graph, denoted as $\mathcal{E}_T^s \in \mathbb{R}^{n^s \times N \times N}$ and $\mathcal{E}_T^t \in \mathbb{R}^{n^t \times N \times N}$ respectively. Instead of leveraging first-order alignment, we introduce second-order statistics, Deep CORAL M_c in (2), to align the sensor correlations across domains. Building upon this idea, we enhance SCA and introduce improved SCA (iSCA) as shown in (7).

$$d_{mn,T}^{st} = M_c(\mathcal{E}_{mn,T}^s, \mathcal{E}_{mn,T}^t),$$

$$\mathcal{L}_{iSCA,T} = \mathbb{E}(d_{mn,T}^{st}) = \frac{\sum_m \sum_n d_{mn,T}^{st}}{mn}. \quad (7)$$

By aligning each edge between domains with the second-order statistical metric and minimizing the distribution discrepancy of sensor correlations, we can achieve more comprehensive alignment for sensor correlations.

B. Sensor Feature Alignment

Next, we focus on improving sensor feature alignment. The sensor contrasting mechanism in SFA is achieved with the averaged sensor features over the training batches in source and target domains, so it encounters the similar challenge with SCA. Specifically, the contrasting-based alignment procedure in (5) relies on first-order statistics, which might not achieve comprehensive alignment when dealing with complex distributions. To overcome this limitation, we enhance SFA and introduce improved SFA (iSFA) by incorporating Deep CORAL into the contrasting process.

In (8), iSFA begins by calculating sensor-level distribution discrepancies using Deep CORAL to take into account second-order statistics. Subsequently, iSFA leverages the sensor-level discrepancies to perform contrasting between domains. In particular, it minimizes the distribution discrepancies between corresponding sensors across domains while maximizes the distribution discrepancies between different sensors across domains. This approach allows for more comprehensive alignment of

sensor features.

$$\mathcal{L}_{iSFA,T} = -\frac{1}{N} \sum_n \log \frac{e^{M_c(Z_{n,T}^s, Z_{n,T}^t)}}{\sum_j e^{M_c(Z_{n,T}^s, Z_{j,T}^t)}}. \quad (8)$$

C. Multi-Graph Alignment

To combine the alignment for sensor correlations and sensor features across all sequential graph, a straightforward approach would involve computing the average alignment losses for these graphs. However, the distributions across domains for different graphs might exhibit various discrepancies. To account for this and enhance domain adaptation, we propose to learn adaptive weights for adjustments, as shown in (9).

$$\mathcal{L}_{Endo} = \sum_T \mathcal{W}_T (\lambda_{SCA} \mathcal{L}_{iSCA,T} + \lambda_{SFA} \mathcal{L}_{iSFA,T}),$$

$$\mathcal{W}_T = M_c([Z_{1,T}^s, \dots, Z_{L,T}^s], [Z_{1,T}^t, \dots, Z_{L,T}^t]), \quad (9)$$

where $[a, \dots, b]$ represents concatenating the elements from a to b . In (9), the learnable weights $\mathcal{W}_T$ are determined by evaluating the distribution discrepancy of the T th graph between domains. Specifically, $\mathcal{W}_T$ will be large when there is a substantial discrepancy in the distribution of the T th graph between the source and target domains. By assigning larger weights to the graphs with larger distribution discrepancies between domains, we prioritize alignment in those graphs. This adaptive weighting scheme allows us to better adapt the model to the evolving distributions between domains. Furthermore, we also introduce λ_{SCA} and λ_{SFA} to balance the effect of iSCA and iSFA. By combining the exo-feature alignment and the improved endo-feature alignment, the improved overall loss function (10) can be obtained as:

$$\min \mathcal{L} = \mathcal{L}_C + \mathcal{L}_{EXO} + \mathcal{L}_{Endo}. \quad (10)$$

V. EXPERIMENT

A. Datasets and Setup

To evaluate the performance of SEA and SEA++ under various MTS scenarios, we conducted experiments on six public datasets: C-MAPSS [52], Opportunity HAR [53], UCI-HAR [54], WISDM [55], Sleep-EDF [56], and OnHW [57].

C-MAPSS describes the degradation process of aircraft engines. The MTS data in C-MAPSS originated from 14 sensors to measure various physical parameters, such as temperature, fan speed, and pressure. The dataset includes four sub-datasets collected under different working conditions and fault modes, where each sub-dataset represents one domain. We processed the sub-datasets following the data preparation in the previous work [58]. In particular, annotations in this dataset represent the remaining useful life (RUL) of engines and it is a regression task for RUL prediction.

Opportunity HAR describes human activities. The MTS data in Opportunity HAR originated from 113 sensors deployed to various locations of the human body, such as hand and leg. After removing three sensors with constant values, 110 sensors were used for evaluation. The dataset was collected

from four subjects, each representing one domain. The data annotations include two levels: 1). locomotion representing the low-level tasks including four categories, sitting, standing, walking, and lying down; 2). gestures representing the high-level tasks including 17 various actions. Following the experimental settings in the previous work [59], we adopted the low-level tasks. As some values in the data were missing, the linear interpolation was adopted to fill in the missing positions. To construct the training dataset, we adopted a sliding window with a size of 128 and an overlapping of 50% following the previous work [59].

UCI-HAR describes human activities. The MTS data in UCI-HAR originated from nine channels, which correspond to various sensors, such as accelerometers and gyroscopes, attached to the human body. The dataset comprises records from 30 subjects, each performing six activities: walking, walking upstairs, walking downstairs, standing, sitting, and lying down. Given the variability across subjects, we treat each subject as a separate domain. Following the previous research [19], we selected ten cross-domain scenarios for analysis. Furthermore, in line with Opportunity HAR, we utilized a window size of 128 to extract samples from the data.

WISDM also describes human activities. The MTS data in WISDM originated from three channels, with accelerometer sensors attached to the human body. The dataset includes records from 36 subjects, each engaging in the same six activities as in UCI-HAR. Similar to UCI-HAR, we consider each subject as a distinct domain, and we select ten cross-domain scenarios in line with previous research [19]. Compared to UCI-HAR, WISDM presents additional challenges due to imbalanced samples across subjects.

Sleep-EDF describes sleep stages. The MTS data in Sleep-EDF originated from six channels, including EEG (Epz-Cz, Pz-Oz), EOG, and EMG. The dataset includes records from 20 subjects, each classified into five sleep stages: wake, N1, N2, N3, and REM [19]. We treat each subject as a separate domain and select ten cross-domain scenarios following prior study [19]. It's noted that each sample is collected over a duration of 30 seconds with a sampling rate of 100 Hz, resulting in a total of 3000 timestamps per sample. To mitigate training time costs, we down-sample each data to a sampling rate of 10 Hz, resulting in 300 timestamps per sample after downsampling.

OnHW describes English characters by online handwriting. The MTS data in OnHW originated from 13 channels, comprising various sensors like accelerometers, gyroscopes, magnetometers, and force sensors integrated into a digital pen to record signals during writing. The dataset includes records from 119 subjects, each tasked with writing the 26 alphabet characters in both lower and upper cases. Consequently, the dataset is divided into three subsets: Lower (26 classes), Upper (26 classes), and Both (52 classes), denoting lower case only, upper case only, and both cases, respectively. Additionally, the data is further divided into writer-dependent and writer-independent. For cross-domain scenarios, we followed the previous work [43] and chose to transfer from right-handed writers to left-handed writers, considering that right-handed writers are typically more

TABLE I
THE COMPARISONS WITH SOTAS IN C-MAPSS (R: RMSE, S: SCORE)

Models (R)	1→2	1→3	1→4	2→1	2→3	2→4	3→1	3→2	3→4	4→1	4→2	4→3	Avg.
Source	28.76	35.28	27.21	21.86	39.80	33.50	37.90	30.33	24.10	33.76	25.31	21.75	29.96
Target	12.93	13.33	14.66	12.36	13.33	14.66	12.36	12.93	14.66	12.36	12.93	13.33	13.32
DDC	36.47	32.58	35.39	33.06	39.57	31.41	31.74	28.74	34.98	33.26	34.70	31.69	33.63
Deep CORAL	18.07	27.48	26.24	15.13	37.04	29.96	31.83	25.50	21.43	26.97	22.71	19.12	25.12
DANN	16.63	21.18	23.08	14.36	27.38	26.87	22.35	20.14	19.76	20.97	20.76	18.52	21.00
BNM	17.88	38.71	37.42	16.36	43.28	35.97	38.04	34.72	27.31	32.28	26.41	22.21	30.88
SDAT	16.22	22.70	24.25	13.64	28.63	25.54	20.71	19.99	18.03	18.73	18.60	17.95	20.42
AdvSKM	16.66	22.12	23.37	14.35	27.07	<u>24.32</u>	28.47	20.98	19.63	19.65	20.35	16.41	21.11
CoDATs	16.27	20.52	22.90	13.97	27.65	26.20	24.92	23.04	19.09	26.39	21.54	15.51	21.50
CtsADA	18.31	29.23	29.49	23.20	54.69	41.60	38.14	24.67	19.86	31.63	25.71	23.97	30.04
CLUDA	23.65	32.32	28.83	21.81	28.07	29.75	25.58	23.48	25.56	22.51	24.18	22.23	25.66
SEA	15.63	19.75	25.81	<u>12.93</u>	<u>27.02</u>	26.37	23.40	<u>19.96</u>	18.70	19.91	17.77	<u>16.35</u>	<u>20.30</u>
SEA++	<u>15.68</u>	<u>19.86</u>	21.90	12.92	26.10	24.20	19.97	19.49	17.58	<u>19.22</u>	<u>18.32</u>	17.18	19.37
Models (S)	1→2	1→3	1→4	2→1	2→3	2→4	3→1	3→2	3→4	4→1	4→2	4→3	Avg.
Source	9341	4751	4699	2340	8335	11018	24892	26878	11006	16318	8771	2047	10866
Target	678	330	1027	241	330	1027	241	678	1027	241	678	330	569
DDC	18658	8931	27913	9265	6262	28098	17789	47520	64619	23674	46423	12191	25945
Deep CORAL	1303	2039	3905	356	6537	5557	12218	14140	2258	5125	4116	887	4870
DANN	1001	1166	2915	420	2107	3595	<u>1533</u>	5622	2403	1632	2429	631	2121
BNM	1706	8090	28630	587	47739	26290	12556	67465	12435	8718	12730	1993	19078
SDAT	986	796	3375	316	2063	3750	1735	4580	1981	1262	2231	624	1975
AdvSKM	1035	1033	<u>2803</u>	404	1796	<u>3264</u>	5775	4745	2270	<u>1101</u>	3275	579	2340
CoDATs	963	1056	2818	369	1893	4122	3811	9112	2368	<u>3664</u>	4311	413	2908
CtsADA	1467	1965	6832	4136	29153	22515	47682	17418	2138	13019	15436	1593	13613
CLUDA	2177	2801	4052	755	<u>1720</u>	4780	1681	3482	2962	1128	3682	1107	2527
SEA	968	860	3121	261	1793	3403	2270	3947	<u>1954</u>	1238	2228	<u>543</u>	1882
SEA++	938	<u>824</u>	2567	<u>291</u>	1717	3123	1467	<u>3574</u>	1557	1083	2164	613	1659

prevalent than left-handed ones. Thus, we have a total of six cases (three alphabet cases with writer-dependent and writer-independent), each involving the transfer from right-handed to left-handed data.

The experiments include three parts, comparisons with state-of-the-art methods, the ablation study, and sensitivity analysis. All experiments were run ten times with average results reported to remove the effect of random initialization. Besides, we set the batch size as 50 and employed Adam as the optimizer with a learning rate of 0.001. Meanwhile, we adopted 30 training epochs for training our model. Furthermore, we built and trained our model based on Pytorch 1.9 and NVIDIA GeForce RTX 3080Ti GPU.

We employed different evaluation indicators to assess the performance of our method on the two distinct tasks. For regression tasks, e.g., RUL prediction, we utilized Root Mean Square Error (RMSE) and the Score function [58], [60], [61], [62] as evaluation metrics. Lower values for both indicators indicate better model performance. For classification tasks, we employed accuracy as the evaluation metric. High accuracy values signify superior model performance.

B. Comparisons With State-of-the-Art

We compared SEA and SEA++ with State-Of-The-Art methods (SOTAs), including general UDA methods and TS UDA methods. General UDA methods were designed to be applicable to various tasks and domains, regardless of the specific data type, including DDC [9], Deep CORAL [13], DANN [10],

BNM [63], LMMD [64], and SDAT [65]. TS UDA methods were specifically designed for TS data, including AdvSKM [41], CoDATs [39], CtsADA [58], SLARDA [59], and CLUDA [66]. For fair comparisons, all methods were run ten times based on the same feature extractor (i.e., GNN-LSTM). Since the target domain labels are unavailable, selecting optimal hyperparameters using target domain samples poses a challenge. Thus, we adopted the source risk approach for hyperparameter optimization, as outlined in previous works [10], [19], [67], to implement all the baseline methods on the six datasets. Specifically, we selected the candidate model that achieves the best evaluation metric, e.g., accuracy for classification and MSE for regression, on a validation set from the source domain. During implementations, LMMD and SLARDA were designed specifically for multi-class classification UDA, so they are not included for RUL prediction on C-MAPSS dataset. Notably, in the previous SEA paper, different cross-domain scenarios of C-MAPSS were based on various sample lengths. For example, the cross-domain scenario 1→2 was based on a time length of 72 while 3→4 was based on a time length of 45. However, this training setting may be inapplicable in real-world systems where obtaining samples with a fixed time length is desired. Thus, results in the C-MAPSS dataset have been re-conducted based on a consistent time length of 60 timestamps. Additionally, we have included the results of Target-only (Target) and Source-only (Source) baselines for comparison. The target-only baseline represents the upper bound of UDA, where training and testing were performed on the target domain only. The source-only baseline represents the lower

TABLE II
THE COMPARISONS WITH SOTAS IN OPPORTUNITY HAR

Models	1→2	1→3	1→4	2→1	2→3	2→4	3→1	3→2	3→4	4→1	4→2	4→3	Avg.
Source	47.50	64.50	60.83	73.00	71.00	65.67	64.83	56.67	72.17	66.67	65.50	74.67	65.25
Target	91.90	97.95	95.60	98.40	97.95	95.60	98.40	91.90	95.60	98.40	91.90	97.95	95.96
DDC	84.40	76.50	84.95	80.95	81.30	78.45	81.05	77.65	83.25	85.65	83.65	76.35	81.18
Deep CORAL	86.30	79.70	84.25	86.85	<u>85.45</u>	83.05	85.05	78.85	86.30	82.45	85.40	90.25	84.49
DANN	83.40	<u>85.30</u>	<u>85.95</u>	87.05	82.85	82.25	86.30	81.10	87.25	85.65	83.60	90.70	85.12
BNM	79.00	80.25	79.10	86.65	80.45	81.75	79.15	79.30	79.85	82.40	82.20	83.90	81.17
LMMD	78.95	82.35	77.10	81.60	82.10	75.95	73.25	72.80	79.50	83.70	78.10	81.90	78.94
SDAT	86.75	84.20	84.35	<u>89.10</u>	83.20	83.25	85.60	79.90	80.95	82.85	80.95	87.85	84.08
AdvSKM	86.05	84.15	80.35	86.45	83.70	80.70	83.40	77.50	86.75	81.50	80.25	86.60	83.12
CoDATs	84.25	82.45	82.95	88.50	85.20	<u>85.40</u>	82.35	80.90	81.70	81.45	84.00	86.55	83.81
CtsADA	80.00	73.67	76.08	69.75	78.17	79.08	67.75	77.33	64.75	74.58	52.17	75.00	72.36
SLARDA	80.38	80.50	79.50	82.75	78.88	79.25	75.88	78.62	70.12	80.62	84.00	77.25	78.98
CLUDA	71.50	75.00	75.00	73.75	74.75	75.50	67.75	72.50	62.75	59.50	68.00	71.00	70.58
SEA	86.85	83.45	85.85	87.35	85.70	83.75	86.90	81.10	89.25	87.30	84.25	91.20	86.08
SEA++	<u>86.30</u>	90.40	88.25	90.25	84.55	87.15	87.30	81.75	89.50	91.75	85.45	92.15	87.90

TABLE III
THE COMPARISONS WITH SOTAS IN UCI-HAR

Models	2→11	6→23	7→13	9→18	12→16	18→27	20→5	24→8	28→27	30→20	Avg.
Source	87.50	73.83	79.38	60.67	65.83	92.83	66.04	80.42	93.50	71.83	77.18
Target	98.12	99.10	97.25	99.80	90.70	99.80	96.88	95.12	99.80	97.80	97.44
DDC	94.62	82.90	87.88	73.50	76.10	95.10	72.75	92.00	95.30	80.30	85.05
Deep CORAL	96.25	85.40	88.25	72.50	73.90	98.00	<u>74.25</u>	89.50	95.50	79.20	85.28
DANN	98.00	<u>87.80</u>	86.38	74.00	69.60	98.70	73.62	94.25	95.00	80.90	85.83
BNM	95.00	83.50	86.12	73.70	73.20	96.80	69.75	86.00	96.60	76.90	83.76
LMMD	94.25	76.00	86.00	60.30	73.00	96.10	65.50	85.25	94.00	74.70	80.51
SDAT	83.88	78.60	75.75	69.40	64.20	94.70	74.25	89.25	89.10	72.20	79.13
AdvSKM	98.25	85.20	89.25	69.70	78.00	96.10	65.88	93.25	95.70	75.90	84.72
CoDATs	98.25	84.30	88.50	73.10	72.00	98.10	72.25	90.01	97.00	<u>83.00</u>	85.65
CtsADA	95.05	82.59	88.69	53.27	61.09	96.02	68.57	68.71	93.45	68.88	77.63
SLARDA	97.79	84.11	89.95	65.55	70.64	100.00	72.64	96.24	97.05	81.68	85.57
CLUDA	96.00	86.50	85.00	62.70	69.10	88.00	72.00	88.40	91.30	78.20	81.72
SEA	99.12	85.50	<u>90.65</u>	75.60	<u>79.30</u>	98.70	73.75	90.25	<u>97.50</u>	81.30	<u>87.17</u>
SEA++	<u>98.39</u>	89.00	90.71	<u>74.14</u>	80.71	<u>98.71</u>	74.50	<u>96.00</u>	98.00	84.86	88.50

bound of UDA, where only the source data were used for training without any UDA methods applied.

Tables I–VI show the results on the cross-domain scenarios of all six datasets. From these results, we observe that both SEA and SEA++ achieve better performance than SOTA methods in most cross-domain scenarios and outperform all the baselines in terms of average performance. For example, SEA made significant improvements on C-MAPSS, UCI-HAR, and Sleep-EDF, improving by 4.7% (Score), 1.34%, and 1.36%, respectively, compared to the best baselines. However, SEA’s lower-order alignment restricts its effectiveness in aligning sensor-level information. Meanwhile, ignoring the alignment of multi-graphs restricts the alignment of evolving sensor-level distributions, limiting SEA’s performance. By addressing these issues, SEA++ further improves performance on all downstream datasets. On C-MAPSS, UCI-HAR, and Sleep-EDF, SEA already achieved significant improvements, while SEA++ further improves by 11.84%, 1.33%, and 1.58% compared to SEA, resulting in improvements of 16%, 2.67%, and 2.94% compared to best baselines. Additionally, SEA++ improves by 1.82%, 2.54%, and 2.25% compared to SEA, and improves by 2.78%, 2.62%, and

2.57% compared best baselines on Opportunity HAR, WISDM, and OnHW respectively.

These results indicate the effectiveness of SEA and SEA++, showing the necessity of aligning sensor-level information. Furthermore, SEA++ addresses the minor limitations of SEA, resulting in improved performance that establishes its state-of-the-art status compared to traditional methods.

C. Ablation Study

To comprehensively evaluate the effectiveness of our endo-feature alignment and the associated improvements, we have conducted extensive ablation studies consisting of nine different variants, which can be categorized into three aspects. The first aspect compares our method with a variant that only performs exo-feature alignment without utilizing any sensor-level alignment. In the second and third aspects, we consider variants combined exo-feature alignment with one of endo-feature alignment techniques: SCA or SFA. Each aspect further includes four variants based on different improvements. We utilize the variants with SCA for illustration. The “w/Exo+SCA” variant represents

TABLE IV
THE COMPARISONS WITH SOTAs IN WISDM

Models	7→18	20→30	35→31	17→23	6→19	2→11	33→12	5→26	28→4	23→32	Avg.
Source	41.60	41.00	36.12	41.67	31.75	36.33	36.00	39.38	41.67	37.17	38.27
Target	81.90	83.60	88.88	68.17	86.25	67.33	82.38	74.00	77.67	65.50	77.57
DDC	69.60	69.20	55.50	62.33	60.83	45.33	70.25	65.25	62.67	47.00	60.80
Deep CORAL	74.20	69.40	<u>58.75</u>	59.33	55.33	47.00	66.00	69.25	69.00	40.67	60.89
DANN	69.00	69.60	<u>54.50</u>	67.00	<u>62.17</u>	47.00	<u>76.25</u>	64.50	69.67	48.00	62.77
BNM	72.80	68.40	57.25	55.33	48.67	45.33	60.00	69.50	68.00	46.33	59.16
LMMD	73.80	66.20	58.25	62.33	50.00	46.33	75.00	<u>70.00</u>	62.00	44.00	60.79
SDAT	74.20	70.20	52.25	60.00	54.83	46.67	72.75	<u>62.75</u>	67.33	48.00	60.90
AdvSKM	49.83	44.33	36.25	48.89	36.94	37.78	35.83	46.46	50.00	54.28	44.06
CoDATs	43.00	41.67	35.83	48.61	30.69	40.00	37.29	50.62	49.44	55.00	43.22
CtsADA	73.77	<u>71.07</u>	53.13	67.00	62.09	45.16	65.98	62.44	66.36	48.70	61.57
SLARDA	74.15	70.10	51.33	64.00	47.88	45.00	53.10	68.05	73.33	48.41	59.54
CLUDA	64.80	60.40	50.40	58.00	53.40	36.44	60.40	54.12	56.40	52.00	54.64
SEA	<u>75.00</u>	69.00	55.25	59.67	59.17	48.33	69.75	68.22	69.12	<u>55.00</u>	62.85
SEA++	75.70	71.90	59.12	<u>64.17</u>	62.17	<u>47.17</u>	76.75	70.25	<u>71.67</u>	55.00	65.39

TABLE V
THE COMPARISONS WITH SOTAs IN SLEEP-EDF

Models	0→11	7→18	9→14	12→5	16→1	3→19	18→12	13→17	5→15	6→2	Avg.
Source	57.47	48.04	64.43	49.46	54.22	48.21	47.67	48.09	62.79	60.17	54.06
Target	84.14	83.47	77.74	76.07	75.12	80.04	75.11	79.80	83.04	82.81	79.73
DDC	55.74	58.27	65.81	58.00	63.76	55.90	57.82	51.33	60.96	75.28	60.29
Deep CORAL	61.82	60.00	67.62	58.16	69.81	62.52	56.88	52.28	65.88	75.32	63.03
DANN	58.82	63.34	67.62	60.88	70.25	61.67	<u>58.38</u>	<u>53.85</u>	69.35	75.00	63.92
BNM	60.98	55.27	66.90	51.86	69.03	60.54	55.98	53.35	64.08	74.92	61.29
LMMD	60.74	55.64	64.28	55.54	66.76	59.71	55.35	50.11	62.34	72.93	60.34
SDAT	<u>63.58</u>	61.88	73.21	57.34	70.82	62.19	57.77	53.05	67.28	71.42	63.85
AdvSKM	61.68	62.34	72.00	58.04	72.04	63.24	55.90	49.98	63.64	75.82	63.47
CoDATs	62.94	61.57	70.45	56.41	68.97	60.08	56.75	47.94	69.65	74.33	62.91
CtsADA	55.99	48.04	63.59	48.58	62.24	53.56	52.02	50.42	56.48	67.95	55.89
SLARDA	56.80	49.93	66.64	53.70	71.08	56.44	54.99	53.43	62.39	64.76	59.02
CLUDA	54.20	56.09	71.91	54.27	70.31	55.31	49.25	41.69	68.12	67.17	58.83
SEA	59.84	<u>65.34</u>	<u>74.00</u>	<u>61.55</u>	<u>72.44</u>	<u>63.67</u>	57.23	56.49	<u>71.91</u>	70.35	<u>65.28</u>
SEA++	65.91	65.89	75.14	63.49	75.04	63.91	60.47	50.63	72.75	<u>75.33</u>	66.86

TABLE VI
THE COMPARISONS WITH SOTAs IN ONHW

Models	Lower dep	Lower indep	Upper dep	Upper indep	Both dep	Both indep	Avg.
Source	48.54	47.37	52.12	48.30	40.82	39.40	46.09
Target	85.55	85.55	82.00	82.00	76.82	76.82	81.46
DDC	66.91	63.73	64.00	62.50	62.91	59.91	63.33
Deep CORAL	70.45	63.64	<u>69.50</u>	66.50	61.45	58.50	65.01
DANN	73.64	66.36	<u>69.50</u>	65.00	62.18	58.18	65.81
BNM	51.82	45.00	53.00	53.50	48.41	50.91	50.44
LMMD	55.64	46.36	53.50	49.50	50.00	46.14	50.19
SDAT	67.73	64.18	73.00	64.50	59.32	54.95	63.95
AdvSKM	63.64	61.64	67.00	61.50	62.73	59.36	62.65
CoDATs	61.82	55.91	59.00	57.50	56.36	53.32	57.32
CtsADA	57.48	51.68	58.61	55.92	51.76	46.55	53.67
SLARDA	59.84	54.58	61.14	56.87	58.78	54.67	57.65
CLUDA	59.10	58.44	59.74	58.89	53.89	51.79	56.98
SEA	68.90	<u>67.95</u>	69.15	<u>67.05</u>	63.82	59.88	<u>66.13</u>
SEA++	<u>72.05</u>	67.55	71.44	68.50	66.89	63.86	68.38

utilizing vanilla SCA for sensor correlation alignment. By introducing Multi-Graph Alignment (MGA), we have the “w/ Exo + SCA + MGA” variant, while still utilizing first-order alignment. By introducing alignment with second-order statistics without MGA, we have the variant “w/ Exo + SCA + 2-order”.

Finally, by incorporating both improvements, i.e., MGA and second-order alignment, we obtain the “w/ Exo + iSCA” variant. The same variations were applied for SFA, resulting in four additional variants. Notably, only the results on C-MAPSS and Opportunity HAR have been reported, and the full results on

TABLE VII
THE ABLATION STUDY IN C-MAPSS (R: RMSE, S: SCORE)

Variants (R)	1→2	1→3	1→4	2→1	2→3	2→4	3→1	3→2	3→4	4→1	4→2	4→3	Avg.
w/ Exo (w/o endo)	18.73	28.12	26.37	15.46	37.30	30.11	32.66	26.01	22.10	27.67	23.15	19.70	25.61
W/ Exo + SCA	16.41	23.76	24.08	14.54	31.80	27.42	31.29	23.49	19.26	26.94	22.01	17.53	23.21
W/ Exo + SCA + MGA	16.10	22.07	23.30	14.15	29.67	27.08	28.53	22.04	18.54	26.91	20.89	17.04	22.19
W/ Exo + SCA + 2-order	16.45	21.61	23.59	13.55	33.16	27.94	25.63	21.40	18.61	23.81	19.85	17.63	21.94
W/ Exo + iSCA	15.65	20.33	22.19	13.31	25.46	24.61	27.98	21.64	18.70	26.15	20.31	17.64	21.16
W/ Exo + SFA	15.07	23.00	27.84	13.71	30.79	30.28	32.22	19.89	20.00	22.15	19.11	17.33	22.62
W/ Exo + SFA + MGA	16.75	22.12	22.88	14.34	28.59	26.92	28.48	21.67	18.56	23.10	19.61	17.13	21.68
W/ Exo + SFA + 2-order	16.38	23.10	23.19	13.97	30.42	27.32	25.91	21.38	18.49	24.88	21.10	18.32	22.04
W/ Exo + iSFA	15.72	21.35	23.09	12.86	27.14	26.78	22.48	19.64	19.35	22.58	18.31	16.17	20.46
SEA++	15.68	19.86	21.90	12.92	26.10	24.20	19.97	19.49	17.58	19.22	18.32	17.18	19.37
Variants (S)	1→2	1→3	1→4	2→1	2→3	2→4	3→1	3→2	3→4	4→1	4→2	4→3	Avg.
w/ Exo (w/o endo)	1515	1974	3951	363	6176	5579	10268	12167	2296	4973	4125	894	4523
W/ Exo + SCA	1183	1621	3181	343	2452	4290	9029	9240	2096	4233	3990	801	3538
W/ Exo + SCA + MGA	1003	989	2950	393	2260	3896	9374	7303	1980	4057	3673	713	3216
W/ Exo + SCA + 2-order	1067	1070	3059	342	3514	4932	5814	8421	1846	2560	3207	752	3049
W/ Exo + iSCA	949	973	2784	320	1609	3169	5511	6609	1974	3441	3459	446	2604
W/ Exo + SFA	1015	1886	3345	305	2006	5621	9756	5958	2270	3166	3225	639	3266
W/ Exo + SFA + MGA	955	1241	2652	397	1884	5724	6193	6046	2131	2457	3071	675	2786
W/ Exo + SFA + 2-order	1026	1191	2892	385	2621	4668	5417	5112	1700	2926	4174	784	2741
W/ Exo + iSFA	1068	1339	3582	275	2179	5266	3547	2678	1968	2556	2276	547	2273
SEA++	938	824	2567	291	1717	3123	1467	3574	1557	1083	2164	613	1659

TABLE VIII
THE ABLATION STUDY IN OPPORTUNITY HAR

Variants (R)	1→2	1→3	1→4	2→1	2→3	2→4	3→1	3→2	3→4	4→1	4→2	4→3	Avg.
w/ Exo (w/o endo)	85.84	78.79	84.03	86.52	83.27	82.97	84.92	77.92	86.28	81.76	84.93	89.99	83.94
W/ Exo + SCA	84.97	83.07	84.36	86.56	83.26	82.88	84.90	78.19	86.63	81.63	84.47	88.45	84.12
W/ Exo + SCA + MGA	85.00	86.75	85.05	88.25	82.60	83.20	85.25	79.75	86.80	83.90	85.85	88.35	85.06
W/ Exo + SCA + 2-order	85.30	86.95	86.55	89.30	82.95	83.05	85.60	78.40	87.35	83.00	84.80	89.90	85.26
W/ Exo + iSCA	86.30	90.00	85.65	90.00	85.20	84.35	85.70	81.00	89.00	85.95	84.20	90.50	86.49
W/ Exo + SFA	85.93	84.40	84.22	87.06	83.74	83.67	84.44	78.44	86.50	82.48	84.47	88.84	84.51
W/ Exo + SFA + MGA	85.95	87.60	85.75	88.05	84.75	83.90	84.60	79.05	87.35	83.50	83.95	89.90	85.36
W/ Exo + SFA + 2-order	85.60	86.95	84.75	87.85	84.25	82.85	84.70	79.30	88.95	84.30	84.10	89.65	85.27
W/ Exo + iSFA	87.45	88.95	87.30	88.50	85.45	84.90	86.60	81.15	89.45	87.30	84.15	91.75	86.91
SEA++	86.30	90.40	88.25	90.25	84.55	87.15	87.30	81.75	89.50	91.75	85.45	92.15	87.90

the remaining datasets can be found in our appendix, available online.

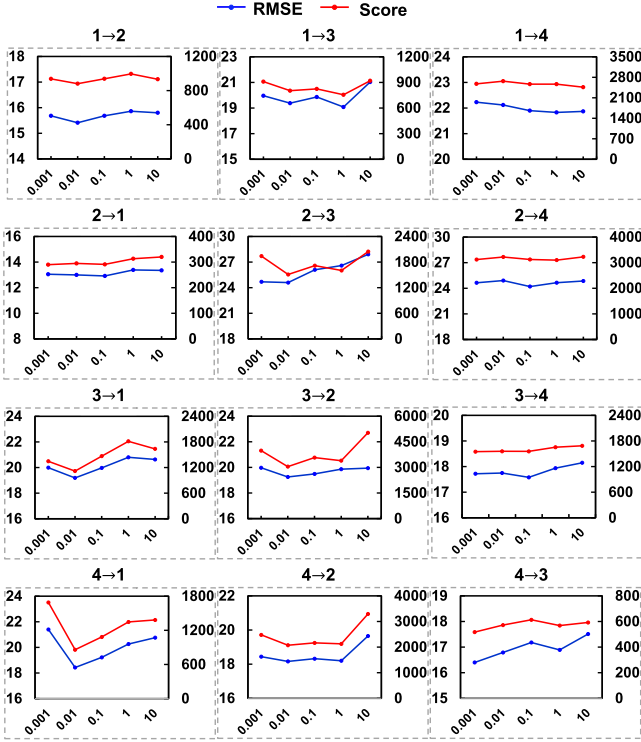
Tables VII and VIII show the ablation study on C-MAPSS and Opportunity HAR respectively. We focus on the average RMSE results on C-MAPSS for illustration and consider the variant “w/ Exo” as the baseline. First, we observe that introducing SCA and SFA leads to improvements of 9.38% and 11.71%, respectively, indicating the effectiveness of endo-feature alignment. As the endo-feature alignment still has minor limitations, we further introduce MGA and second-order statistic alignment to improve the sensor-level alignment. Introducing MGA leads to performance improvements of 13.36% (SCA) and 15.37% (SFA), while introducing second-order statistic alignment improves by 14.36% (SCA) and 13.96% (SFA). The improvements indicate that MGA and second-order statistic alignment both show effectiveness for better sensor-level alignment, so it’s reasonable to combine both modules to further improve performance. Combining both parts, the variants “w/ iSCA” and “w/ iSFA” improve the performance by 17.37% and 20.14% respectively compared to the baseline. Finally, by combining

iSCA and iSFA for complete endo-feature alignment, we achieve a remarkable improvement of 24.39% compared to the baseline. Similar trends can also be observed in Opportunity HAR, as shown in Table VIII. Regarding the average accuracy, SEA++ increases by 3.96% compared to the model with exo-feature alignment only. The results in the cross-domain scenarios of Opportunity HAR can further verify the effectiveness of our proposed modules.

These experiments highlight the significance of endo-feature alignment in achieving effective alignment at the sensor level for MTS-UDA. Furthermore, the additional modules for endo-feature alignment, namely MGA and higher-order statistics alignment, prove to be highly effective in further improving the performance of MTS-UDA.

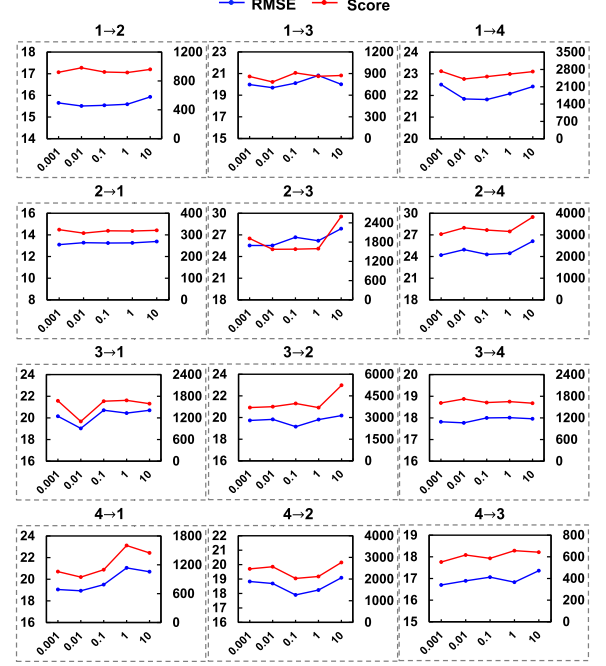
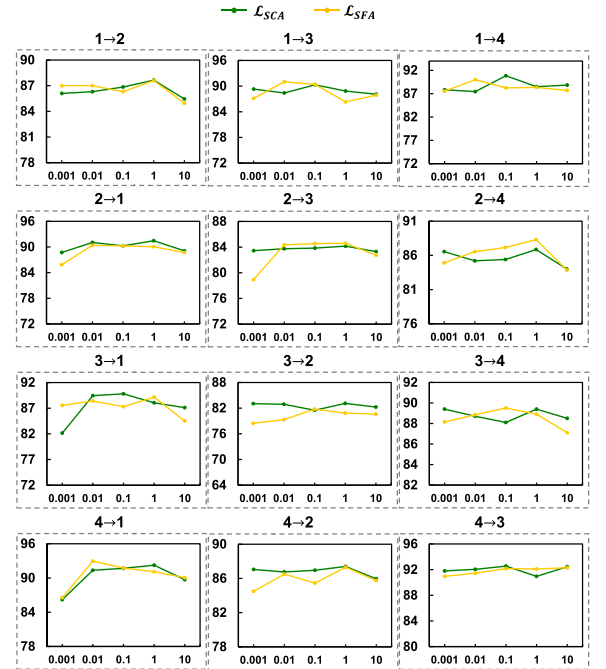
D. Sensitivity Analysis

We have conducted the sensitivity analysis to evaluate the impact of hyperparameters on our model’s performance, including four hyperparameters: λ_{SCA} , λ_{SFA} , number of heads,

Fig. 5. The sensitivity analysis for λ_{SCA} on C-MAPSS.

and patch sizes. λ_{SCA} and λ_{SFA} control the balance between sensor correlation alignment and sensor feature alignment. The number of heads determines the number of graph branches used to model spatial dependencies between sensors. The patch sizes determine the number of sequential graphs constructed. Conducting sensitivity analysis on these hyperparameters helps determine their sensitivity and find optimal values. Notably, the sensitivity analysis presented here pertains specifically to SEA++. Additional details regarding the analysis related to SEA can be found in our appendix, available online.

1) *Analysis for λ_{SCA} and λ_{SFA} :* We have conducted experiments with various values of λ_{SCA} and λ_{SFA} varying from 0.001 to 10 with an interval of $10\times$. Figs. 5 and 6 present the analysis of λ_{SCA} and λ_{SFA} on C-MAPSS respectively, and Fig. 7 shows the analysis for λ_{SCA} and λ_{SFA} on Opportunity HAR. From the results, we can find similar trends for the two hyperparameters, which include two points. First, our method demonstrates robustness to changes in hyperparameters in most cases. For example, we observe insensitivity to λ_{SCA} in cross-domain scenarios such as 1→4, 2→1, and 2→4 of C-MAPSS, as well as 1→3, 2→3, and 4→2 of Opportunity HAR. Similarly, regarding λ_{SFA} , we observe insensitivity in scenarios 1→2, 2→1, and 3→4 of C-MAPSS, as well as 1→4, 3→2, and 4→3 of Opportunity HAR. These results indicate that our method can maintain stable performance without requiring extensive hyperparameter tuning, making it applicable for real-world scenarios. Second, in cases where our method is sensitive to hyperparameters, optimal performance is typically achieved when their values fall within the range of [0.01, 1]. Taking the

Fig. 6. The sensitivity analysis for λ_{SFA} on C-MAPSS.Fig. 7. The sensitivity analysis for λ_{SCA} and λ_{SFA} on Opportunity HAR.

results on C-MAPSS as examples, we observe that our method achieves poor performance when λ_{SCA} is set to 10 in the case of 3→2, and when λ_{SCA} is set to 0.001 in the case of 4→1. Similar trends can also be observed for λ_{SFA} , e.g., $\lambda_{SFA} = 10$ in 2→3. Based on these observations, we recommend setting the values of these hyperparameters within the range of [0.01, 1], where our method consistently achieves good performance.

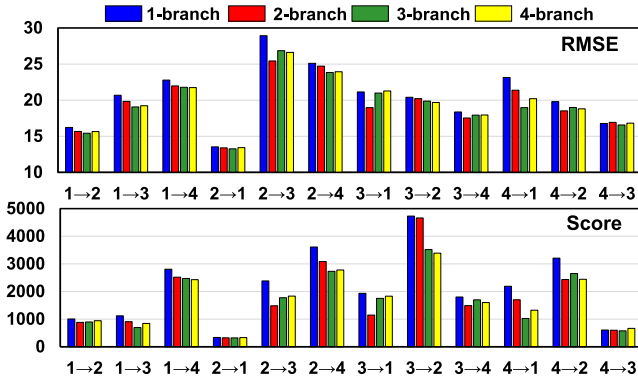


Fig. 8. The sensitivity analysis for different heads on C-MAPSS.

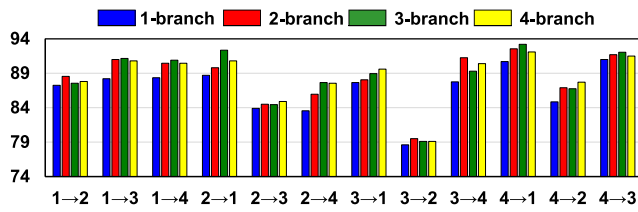


Fig. 9. The sensitivity analysis for different heads on Opportunity HAR.

2) *Analysis for the Number of Heads:* For constructing a graph with MSGC, we have multiple branches with differently initialized weights to model sensor correlations in various feature spaces. In this part, we aim to evaluate the effect of the number of branches on the model performance. Figs. 8 and 9 show the results on C-MAPSS and Opportunity HAR respectively. In our analysis, we analyze the effect using a maximum of 4 branches, as we observe that further increasing the number of branches does not yield significant improvements. The results consistently show that SEA++ obtains performance improvements with increasing the number of branches. Specifically, SEA++ with only one branch tends to exhibit poorer performance in most cases, such as 2→3 of C-MAPSS and 2→4 of Opportunity HAR. With increasing the number of branches, the performance improves. It's worth noting that the improvements become less pronounced when the number of branches is large enough, particularly when transitioning from 3 to 4 branches. This suggests that using 3 branches is generally sufficient for effectively capturing the spatial dependencies within MTS data. Overall, the analysis demonstrates the importance of leveraging multiple branches in MSGC for modeling the complex relationships among sensors. Increasing the number of branches allows our method to capture a more comprehensive understanding of sensor correlations, leading to improved performance for MTS-UDA. Meanwhile, three branches are good enough to capture such relational information.

3) *Analysis for Patch Size:* We segment each MTS sample into multiple patches with a predefined size d , construct sequential graphs with the patches, and conduct multi-graph alignment to align the evolving distributions within MTS data. The patch size d affects sequential graph construction and the

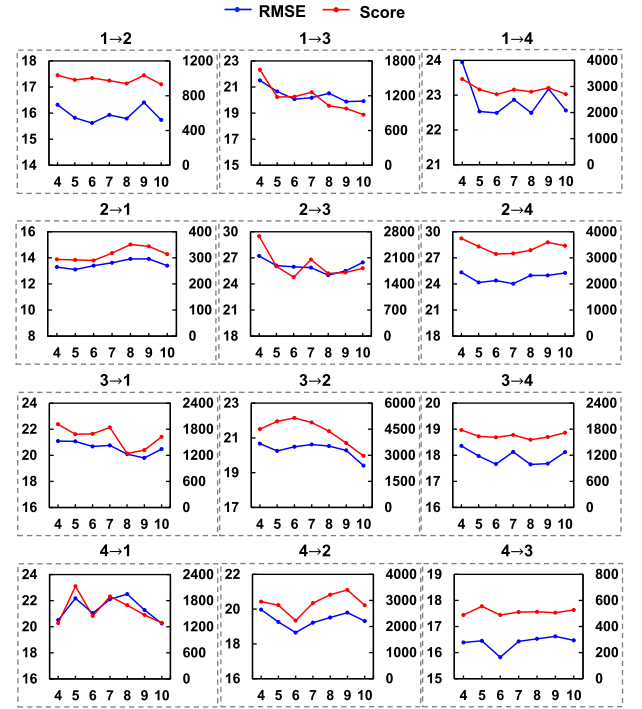


Fig. 10. The sensitivity analysis for patch sizes on C-MAPSS.

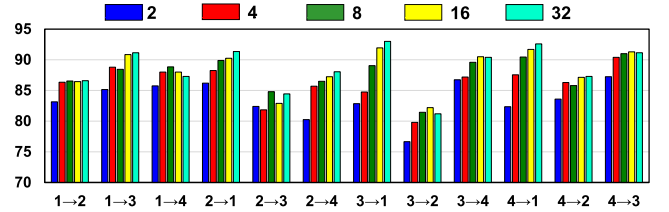


Fig. 11. The sensitivity analysis for patch sizes on Opportunity HAR.

alignment of evolving distributions, so evaluating its effects is important. Notably, we conducted the analysis with different patch size choices for C-MAPSS and Opportunity HAR, as their samples have different lengths. For C-MAPSS, the time length of each sample is fixed as 60, so we use patch sizes within [4,5,6,7,8,9,10] for patch size analysis. For Opportunity HAR, the time length of each sample is fixed as 128, so we use patch sizes within [2,4,8,16,32].

Based on the analysis of C-MAPSS in Fig. 10 and Opportunity in Fig. 11, it is evident that increasing patch sizes tends to result in better performance. This observation holds true in several cross-scenarios, such as 2→3, 3→1, and 3→4 of C-MAPSS, where the best performance is achieved with patch sizes of 8 or 9. Similar observations can be found in Opportunity HAR, such as 1→3, 2→1, and 3→1, where the best performance is achieved with patch sizes of 16 or 32. Conversely, large patch sizes can also result in diminished performance. For instance, in cross-domain scenarios 2→3 and 3→1 of C-MAPSS, setting a patch size of 10 yields decreased performance. Similarly, in scenarios 1→4 and 3→2 of Opportunity HAR, employing a patch size of 32 adversely affects performance. The reason behind these trends

lies in two factors. First, we construct graphs to capture spatial dependencies by segmenting MTS data into patches. During graph construction, small patch sizes may not contain enough information to fully capture the complex spatial dependencies among sensors for each patch, affecting the effective transfer of the spatial information across domains. Second, large patch sizes would reduce the number of sequential graphs, thereby affecting the ability to effectively capture the distributions in local temporal intervals and consequently impacting the alignment of evolving distributions across domains. Therefore, it is recommended to make trade-off by choosing suitable patch sizes for better MTS-UDA.

VI. CONCLUSION

In this paper, we formulate Multivariate Time-Series Unsupervised Domain Adaptation (MTS-UDA). By analyzing the problems underlying this task, we propose SEnsor Alignment (SEA) to address these issues. To reduce the domain discrepancy at both the local and global sensor levels, we design endo-feature alignment and exo-feature alignment. At the local sensor level, endo-feature alignment aligns sensor features and sensor correlations across domains, preventing misalignment at the sensor level. To enhance the endo-feature alignment, we further design SEA++, incorporating higher-order statistic alignment and a multi-graph alignment technique. These enhancements are specifically designed to facilitate comprehensive alignment across domains and effectively capture the evolving data distributions within MTS data. At the global sensor level, we enforce restrictions on global sensor features to reduce domain discrepancy. Our extensive experiments in all the six public datasets demonstrate the effectiveness of SEA and SEA++ for MTS-UDA.

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